

LAB EXPERIMENT – PART 2

Performance Analysis Using Type-2 Hypervisor

VMware Workstation

CPU Performance Analysis of an Ubuntu Virtual Machine using Sysbench

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1. Aim

To create and verify a virtual machine on the VMware Workstation Type-2 hypervisor, inspect its CPU, memory and disk configuration, monitor resource utilization, and measure CPU performance using Sysbench.

2. Experimental Configuration

Configured / Observed Value	Parameter
VMware Workstation (Type-2)	Hypervisor
Ubuntu 22.04.5 LTS	Guest OS
1 processor, 2 cores = 2 vCPU	CPU allocation
13th Gen Intel Core i5-13450HX presented to guest	CPU model
2048 MB (approximately 2 GB)	Memory allocation
20 GB virtual disk	Virtual disk
NAT	Network configuration
VMware / full virtualization	Virtualization information

3. Procedure

- Launch VMware Workstation and select Create a New Virtual Machine.
- Choose the Typical configuration and select the Ubuntu ISO image.
- Create the VM and configure a 20 GB virtual disk.
- Customize hardware to allocate 2 GB RAM and 2 vCPU; configure the network adapter using NAT.
- Power on the VM, install Ubuntu, restart, and log in.
- Verify the VM using `hostnamectl`, `lscpu`, `free -h` and `df -h`.
- Run `top` to observe CPU utilization, memory utilization, processes and load average.
- Install Sysbench and execute `sysbench cpu --cpu-max-prime=20000 run`.
- Record execution time, total events, events per second and latency statistics.
- Shut down the Ubuntu VM using `sudo poweroff` after completing the measurements.

4. Commands Used

```
hostnamectl
lscpu
free -h
df -h
top
sudo apt update
sudo apt install sysbench -y
sysbench --version
sysbench cpu --cpu-max-prime=20000 run
sudo poweroff
```

5. Observed Results

Observed Result	Metric
Ubuntu 22.04.5 LTS	Operating system
x86_64	Architecture
2 CPUs / 2 cores	Virtual CPUs
13th Gen Intel Core i5-13450HX	CPU model reported by guest
VMware; full virtualization	Virtualization
Approximately 1.9 GiB total	Memory reported by free -h
20 GB virtual disk	Disk
10.0004 s	Sysbench execution time
14,410	Sysbench total events
1,440.80	Sysbench events per second
0.65 ms	Minimum latency
0.69 ms	Average latency
1.89 ms	Maximum latency

6. Resource Utilization Observations

- The guest reports 2 online CPUs and approximately 2 GB of allocated memory.
- The free -h output shows approximately 1.9 GiB total memory, with available memory around 865 MiB at the captured instant.
- The df -h output shows a 20 GB virtual root filesystem, with approximately 12 GB used and 6.5 GB available.
- The top display provides live CPU, memory, process and load information inside the Ubuntu guest.
- Because VMware Workstation is a Type-2 hypervisor, the guest's virtual resources are mediated through the host operating system.

7. Result and Conclusion

Part 2 was completed using VMware Workstation as a Type-2 hypervisor. The Ubuntu virtual machine was configured with 2 vCPUs, 2 GB RAM and a 20 GB virtual disk. The VM configuration was verified using standard Linux system commands, and guest resource utilization was observed using top.

For the supplied Sysbench CPU run, the VM completed the benchmark in 10.0004 seconds and processed 14,410 events at 1,440.80 events per second, with an average latency of 0.69 ms. These values form the Type-2 result set for comparison with the Type-1 Proxmox VE experiment.

8. Type-2 Hypervisor Interpretation

- Architecture: VMware Workstation operates above the host operating system, while Ubuntu runs as the guest OS.
- Host–guest interaction: CPU, memory, storage and networking are presented to Ubuntu as virtual resources and are mediated by the virtualization layer and host.
- Isolation: The guest operates on a virtual disk and virtual CPU resources rather than directly controlling the host's physical devices.
- Typical use cases: Type-2 virtualization is suitable for desktop virtualization, development/testing, laboratory work and running another operating system on a workstation.
- Performance considerations: Benchmark throughput and latency can be affected by virtualization overhead, host workload, resource contention and the configuration of the virtual machine.

9. Final Observation Record

Part 2 Result	Parameter
VMware Workstation	Hypervisor
Type-2	Hypervisor type
Ubuntu 22.04.5 LTS	Guest OS
2 vCPU	CPU
2 GB	Memory
20 GB	Disk
10.0004 s	Total execution time
14,410	Total events
1,440.80	Events per second
0.69 ms	Average latency

10. Screenshots / Experimental Evidence

The supplied screenshots are included below in the same evidence-oriented format as the Type-1 reference. Each figure is placed before its caption, with the caption immediately beneath the corresponding image.

```
soumya@soumya-virtual-machine:~$ hostnamectl
Static hostname: soumya-virtual-machine
Icon name: computer-vm
Chassis: vm
Machine ID: cd97c35e3eb84f5c87658068c9be1f76
Boot ID: af80a014b7bc4c03810e21cd80f56e89
Virtualization: vmware
Operating System: Ubuntu 22.04.5 LTS
Kernel: Linux 6.8.0-138-generic
Architecture: x86-64
Hardware Vendor: VMware, Inc.
Hardware Model: VMware Virtual Platform
```

Figure 1. Ubuntu VM: hostnamectl output showing the hostname, Ubuntu operating system, kernel, architecture and VMware virtual hardware information.

```
soumya@soumya-virtual-machine:~$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 45 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 2
On-line CPU(s) list: 0,1
Vendor ID: AuthenticAMD
Model name: AMD Ryzen 5 5500U with Radeon Graphics
CPU family: 23
Model: 104
Thread(s) per core: 1
Core(s) per socket: 2
Socket(s): 1
Stepping: 1
BogoMIPS: 4191.99
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat p
se36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1g
b rdtscp lm constant_tsc rep_good nopl tsc_reliable nonstop_tsc cpu
id extd_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16 sse4_1 s
se4_2 x2apic movbe popcnt aes xsave avx f16c rdrand hypervisor lahf
_lm cmp_legacy cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw
topoext ssbd ibpb vmcall fsgsbase tsc_adjust bmi1 avx2 smep bmi2 r
dseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 clzer
o arat umip rdpid overflow_recover succor

Virtualization features:
Hypervisor vendor: VMware
Virtualization type: full
Caches (sum of all):
L1d: 64 KiB (2 instances)
L1i: 64 KiB (2 instances)
L2: 1 MiB (2 instances)
L3: 4 MiB (1 instance)
NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0,1
Vulnerabilities:
Gather data sampling: Not affected
Indirect target selection: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Reg file data sampling: Not affected
Retbleed: Mitigation: untrained return thunk; SMT disabled
```

Figure 2. Ubuntu VM: Lscpu output showing x86_64 architecture, 2 CPUs/cores, the presented Intel CPU model and VMware full virtualization.

```
soumya@soumya-virtual-machine:~$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	1.9Gi	1.0Gi	89Mi	41Mi	794Mi	689Mi
Swap:	2.1Gi	3.0Mi	2.1Gi			

Figure 3. Ubuntu VM: free -h output showing memory and swap allocation/usage.

```
soumya@soumya-virtual-machine:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           193M  1.8M  191M   1% /run
/dev/sda3       20G   12G   6.6G  64% /
tmpfs           961M    0  961M   0% /dev/shm
tmpfs           5.0M  4.0K  5.0M   1% /run/lock
/dev/sda2       512M   6.1M  506M   2% /boot/efi
tmpfs           193M  168K  192M   1% /run/user/1000
/dev/sr0        155M  155M    0 100% /media/soumya/CDROM
```

Figure 4. Ubuntu VM: df -h output showing the 20 GB virtual filesystem, used space and available space.

```
soumya@soumya-virtual-machine:~$ top
```

top - 18:46:57 up 2 min, 1 user, load average: 1.07, 0.93, 0.40
Tasks: 340 total, 2 running, 338 sleeping, 0 stopped, 0 zombie
%Cpu(s): 7.0 us, 18.6 sy, 0.0 ni, 74.4 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 1920.3 total, 88.4 free, 1036.8 used, 795.1 buff/cache
MiB Swap: 2140.0 total, 2136.5 free, 3.5 used, 688.0 avail Mem

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1763	soumya	20	0	4349348	257308	129524	R	13.6	13.1	0:22.90	gnome-shell
53	root	0	-20	0	0	0	I	4.5	0.0	0:00.30	kworker/0:1H-kblockd
254	root	20	0	0	0	0	I	4.5	0.0	0:00.55	kworker/u256:26-events_+
363	root	-51	0	0	0	0	S	4.5	0.0	0:00.77	irq/16-vmwgfx
552	systemd+	20	0	14836	6876	6040	S	4.5	0.3	0:00.31	systemd-oomd
2765	soumya	20	0	13392	4364	3364	R	4.5	0.2	0:00.08	top
1	root	20	0	101156	11932	8372	S	0.0	0.6	0:03.18	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.03	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_g
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_p
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
8	root	20	0	0	0	0	I	0.0	0.0	0:00.05	kworker/0:0-events
9	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_hig+
10	root	20	0	0	0	0	I	0.0	0.0	0:00.25	kworker/0:1-cgroup_free
11	root	20	0	0	0	0	I	0.0	0.0	0:02.95	kworker/u256:0-events_f+
12	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_pe
13	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_kthread
14	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_rude_kthread
15	root	20	0	0	0	0	I	0.0	0.0	0:00.00	rcu_tasks_trace_kthread
16	root	20	0	0	0	0	S	0.0	0.0	0:00.20	ksoftirqd/0
17	root	20	0	0	0	0	I	0.0	0.0	0:01.40	rcu_preempt
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.00	migration/0
19	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
20	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
21	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
22	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
23	root	rt	0	0	0	0	S	0.0	0.0	0:00.72	migration/1
24	root	20	0	0	0	0	S	0.0	0.0	0:00.10	ksoftirqd/1
25	root	20	0	0	0	0	I	0.0	0.0	0:00.08	kworker/1:0-rcu_par_gp
26	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/1:0H-kblockd
27	root	20	0	0	0	0	S	0.0	0.0	0:00.01	kdevtmpfs
28	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-inet_
29	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kauditd
30	root	20	0	0	0	0	I	0.0	0.0	0:00.18	kworker/0:2-events

Figure 5. Ubuntu VM: top output showing CPU utilization, memory utilization, load average and active processes.

```
soumya@soumya-virtual-machine:~$ sudo apt update
```

[sudo] password for soumya:
Get:1 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Get:4 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:5 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3,538 kB]
Get:6 http://in.archive.ubuntu.com/ubuntu jammy-backports/main i386 Packages [61.7 kB]
Get:7 http://in.archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [70.2 kB]
Get:8 http://in.archive.ubuntu.com/ubuntu jammy-backports/universe i386 Packages [19.6 kB]
Get:9 http://in.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [30.8 kB]
Get:10 http://in.archive.ubuntu.com/ubuntu jammy-backports/universe Translation-en [16.9 kB]
Get:11 http://security.ubuntu.com/ubuntu jammy-security/main i386 Packages [902 kB]
Get:12 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1,048 kB]
Get:13 http://security.ubuntu.com/ubuntu jammy-security/universe i386 Packages [708 kB]
Fetched 6,652 kB in 8s (811 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
460 packages can be upgraded. Run 'apt list --upgradable' to see them.

Figure 6. Ubuntu VM: Sysbench installation output showing package installation and configuration.

```

soumya@soumya-virtual-machine:~$ sysbench --version
sysbench 1.0.20
soumya@soumya-virtual-machine:~$ sysbench cpu --cpu-max-prime=20000 run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time

Prime numbers limit: 20000

Initializing worker threads...

Threads started!

CPU speed:
  events per second:   636.80

General statistics:
  total time:          10.0016s
  total number of events: 6370

Latency (ms):
  min:                 1.31
  avg:                 1.57
  max:                 6.84
  95th percentile:    2.07
  sum:                 9981.90

Threads fairness:
  events (avg/stddev): 6370.0000/0.00
  execution time (avg/stddev): 9.9819/0.00

```

Figure 7. Ubuntu VM: Sysbench CPU benchmark output for --cpu-max-prime=20000, showing execution time, events, throughput and latency.

11. Performance Analysis Summary

The final benchmark screenshot records 1,440.80 events per second, 14,410 total events and 10.0004 seconds of execution time. Minimum, average and maximum latency are 0.65 ms, 0.69 ms and 1.89 ms respectively.

12. Key Observations

- The guest successfully recognizes VMware as its virtualization environment.
- The VM exposes 2 virtual CPUs and approximately 2 GB of memory to Ubuntu.
- The guest sees a 20 GB virtual disk through the virtual storage layer.
- The Type-2 arrangement introduces an additional software layer between the guest and physical hardware, which can influence performance.
- The Sysbench result provides a reproducible CPU-performance measurement for comparison with the Type-1 result.